Attention-Free Spiking Neural Networks for Efficient Sequence Representation

Abstract

Spiking Neural Networks (SNNs) provide an energy-efficient alternative to conventional Deep Neural Networks by utilizing discrete, event-driven temporal spikes. However, adopting Transformer-based paradigms into the spiking domain typically requires mapping complex Self-Attention mechanisms into spike operations. In this paper, we present an in-depth mathematical formulation and empirical ablation of a Spike-Driven Self-Attention layer—modeled around a Spike-Overlap-like spike intersection metric—to evaluate its true utility. Through comprehensive knowledge distillation, we demonstrate that incorporating Spike-Overlap Attention yields a severe execution bottleneck, quadrupling training time and doubling inference latency (despite having $\sim 3\times$ more parameters), while offering a negligible absolute accuracy gain (+0.0048 Pearson correlation). We propose a purely recurrent, attention-free Spiking Sentence Embedder operating exclusively through Leaky Integrate-and-Fire (LIF) pooling and Backpropagation Through Time (BPTT). Our minimalist architecture incorporates SNN-native hyperpolarization for sequence padding. By explicitly mapping discrete gradients to continuous dense Teacher targets, our model achieves a highly competitive Pearson correlation of 0.8031 on the ALL-STS benchmark. Ultimately, replacing complex spatial attention routing with pure recurrent temporal pooling drastically reduces computational overhead, demonstrating that spatial routing via self-attention provides marginal benefit relative to its cost for neuromorphic sequence representation tasks.

1. Introduction

The computational overhead of Dense Sentence Embedders [1], such as SBERT [2], is heavily dominated by the dense matrix multiplications required in the standard Scaled Dot-Product Attention introduced by Vaswani et al. [3]. While numerous efficient Transformer architectures have been proposed to mitigate this quadratic $\mathcal{O}(L^2)$ complexity [4], they still rely fundamentally on continuous floating-point operations. Spiking Neural Networks (SNNs) alleviate this constraint by operating on sparse, event-driven additions rather than complex multiplications, making them highly suitable for highly parallel neuromorphic hardware [5]. Recent neuromorphic NLP research has focused on translating Self-Attention to SNNs [6, 7] and extending these architectures to extract semantic embeddings. Specific attention mechanisms, such as the Sparse Coincidence-Based Semantic Attention [8], rely heavily on spike overlaps. Furthermore, the development of specialized frameworks like SpikingJelly [9] has accelerated the validation of surrogate gradient methods in deep architectures. However, a critical bottleneck persists: forcibly mapping continuous $\mathcal{O}(L^2)$ spatial attention mechanisms into the discrete spiking domain drastically inflates computational overhead, largely negating the inherent efficiency of SNNs. This raises a fundamental architectural question: *Are explicit spatial attention mechanisms truly necessary for sequence representation, or can the native temporal recurrence of SNNs suffice?*

In this study, we systematically deconstruct the forward dynamics and learning mechanisms of an SNN sequence embedder. We mathematically formulate a Spike-Overlap-based Attention algorithm and utilize it as a comparative baseline to expose the computational inefficiencies of spatial routing. We contrast this against our proposed attention-free Recurrent Pooler. Our empirical findings definitively prove that forcing spatial attention onto SNNs is computationally unjustifiable for this task. Instead, the native temporal integration capabilities of the recurrent pooler, when optimized via Backpropagation Through Time (BPTT) and Surrogate Gradients, are overwhelmingly sufficient for encoding complex semantic structures.

Contributions

Our primary contribution in this study is the proposal and rigorous evaluation of an **Attention-Free Recurrent Spiking Sentence Embedder**. By bypassing complex $\mathcal{O}(L^2)$ spatial routing mechanisms and relying exclusively on temporal Leaky Integrate-and-Fire (LIF) pooling optimized via Backpropagation Through Time (BPTT), we demonstrate that temporal integration is overwhelmingly sufficient for semantic encoding. This minimalist architecture achieves comparable semantic fidelity to spatial attention baselines while offering a 4× reduction in training cost and a 2× reduction in inference latency.

To support and evaluate this primary architecture, we implemented several supporting methodologies:

- A mathematical formulation of Spike-Driven Spike-Overlap Attention, which serves purely as a comparative baseline to expose the computational bottlenecks of spatial routing in SNNs.
- **Hyperpolarized Padding**, an SNN-native sequence padding mechanism that forces padding tokens into a negative refractory state to prevent noise accumulation.
- **Add-Only Delta BPTT routing**, a gradient routing technique that simplifies computations and significantly reduces floating-point matrix multiplications during the backward pass.

2. Methodology: Architectural Dynamics and BPTT

Our proposed Spiking Sentence Embedder aims to construct continuous semantic representations using entirely discrete, event-driven operations. The pipeline consists of Spike Encoding, a comparative Attention vs. Pooling mechanism, and Knowledge Distillation.

2.1. Spike Encoding and Heterogeneous Dynamics

Raw text is tokenized into vocabulary indices. The Spiking Embedding Layer translates these tokens into temporal spike trains $\boldsymbol{S}_{emb}^{(t)}$ over discrete sequence steps $t \in [1, L]$, where the temporal dimension maps perfectly to sequence length. The membrane potential $\boldsymbol{V}_{emb}^{(t)}$ is modeled via Leaky Integrate-and-Fire (LIF) dynamics:

$$\boldsymbol{V}_{emb}^{(t)} = \min\left(1.0, \boldsymbol{\beta}_{emb} \odot \boldsymbol{V}_{emb}^{(t-1)} + \boldsymbol{X}^{(t)}\boldsymbol{W}_{emb}\right) - \boldsymbol{S}_{emb}^{(t)} \odot \boldsymbol{\theta}_{emb} \tag{1}$$

When the pre-reset potential exceeds the threshold $\boldsymbol{\theta}_{emb}$, a spike $\boldsymbol{S}_{emb}^{(t)} = 1$ is fired. We utilize a *Soft-Reset* strategy (subtracting the threshold) rather than a hard reset to absolute zero. Preserving these fractional sub-threshold residual potentials prevents catastrophic semantic information loss across time. Furthermore, we introduce biological heterogeneity: the leak decay $\boldsymbol{\beta}_{emb}$ and thresholds $\boldsymbol{\theta}_{emb}$ are initialized with unique, uniformly distributed values across dimensions (e.g., $\beta \in [1 - 2^{-2}, 1 - 2^{-7}]$). This forces multi-timescale temporal processing immediately upon sequence ingestion.

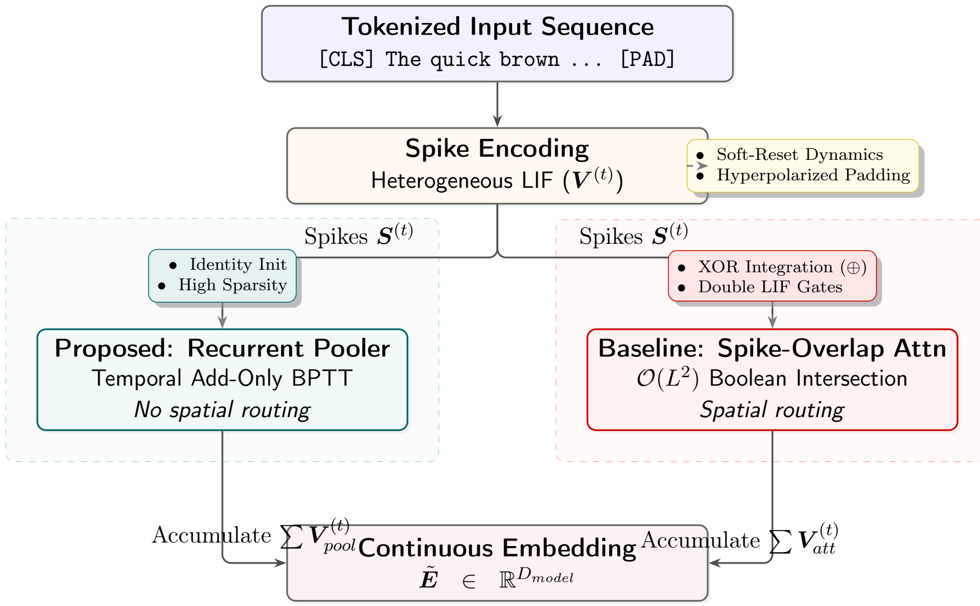

Figure 1: Architectural flowchart contrasting the proposed minimalist Recurrent Pooler against the $\mathcal{O}(L^2)$ Spike-Overlap Attention baseline. The Recurrent Pooler completely bypasses quadratic spatial routing, natively translating discrete token spikes into a continuous temporal embedding state.

Sequence Padding and Hyperpolarization: To handle variable sequence lengths efficiently without relying on explicit attention padding masks, we introduce an SNN-native hyperpolarization technique. The embedding weights associated with the ‘<PAD>‘ token (index 0) are explicitly overridden to -1.0. Consequently, during padding time-steps, the incoming synaptic dot product yields a strongly negative current, hyperpolarizing the membrane potential ($V < 0$). This guarantees that padding tokens emit exactly zero spikes, completely eliminating noise accumulation during temporal pooling.

2.2. Ablation Baseline: Spike-Driven Spike-Overlap Attention

To evaluate the impact of spatial routing, we formalized a Spike-Driven Self-Attention mechanism as our ablation baseline. Unlike floating-point dot products, this method computes affinities purely through Boolean spike intersections (Spike-Overlap index numerator). Embedding spikes $\boldsymbol{S}_{emb}$ are projected into Query (Q), Key (K), and Value (V) spikes using independent LIF neurons. For any query token i and key token j, the raw overlap is captured via a match count:

$$\boldsymbol{M}_{i,j}^{(t)} = \sum_{d=1}^{D_{model}} \left(\boldsymbol{Q}_{i,d}^{(t)} \wedge \boldsymbol{K}_{j,d}^{(t)} \right) \tag{2}$$

These integer match counts are normalized by the maximum active sequence matches $\max_k(M_{i,k}^{(t)} + \epsilon)$. To maintain the purely spiking pipeline, these normalized fractional scores are passed into an auxiliary set of LIF neurons to generate attention mask spikes $\boldsymbol{S}_{scores}$, which finally gate the Value spikes V via accumulation. While this avoids continuous floating-point multiplications, calculating $\mathcal{O}(L^2 \cdot D_{model})$ Boolean intersections at every single timestep, followed by dual-layered LIF dynamics, imposes severe computational bottlenecks and dramatically inflates execution time.

Spike-Difference (XOR) Integration: Following the generation of the attention-gated spikes (denoted as $\boldsymbol{S}_{att}^{(t)}$), we integrate these contextual features with the original embedding spikes $\boldsymbol{S}_{emb}^{(t)}$ using an exclusive-OR (XOR) logical operation:

$$\boldsymbol{S}_{agg}^{(t)} = \boldsymbol{S}_{emb}^{(t)} \oplus \boldsymbol{S}_{att}^{(t)} \tag{3}$$

Unlike traditional additive residual connections, this XOR integration acts as an "innovation signal." It isolates the feature discrepancy between the local token representation and the global contextual attention. By neutralizing redundant spikes that are active in both the local embedding and global context, the subsequent recurrent pooler is forced to process strictly the novel semantic shifts introduced by the attention layer. This integration is explicitly applied in all our attention-based evaluations.

2.3. The Superior Recurrent Pooler (Proposed Architecture)

We propose that spatial attention is functionally redundant. In our purely recurrent architecture, $\boldsymbol{S}_{emb}^{(t)}$ bypasses attention and directly feeds into a Spiking Dense BPTT Pooler layer. Because the input spike $\boldsymbol{S}_{emb}$ strictly guarantees binary values $\{0, 1\}$, the dense matrix projection is simplified to conditional accumulations:

$$\boldsymbol{X}_{pool}^{(t)} = \sum_{d=1}^{D_{in}} \boldsymbol{W}_d \cdot \mathbb{I}\left[S_{emb,d}^{(t)} = 1 \right] \tag{4}$$

By bypassing floating-point multiplications entirely, this step offers massive computational sparsity. The final continuous sentence embedding E is extracted by accumulating the fractional membrane potentials across the entire

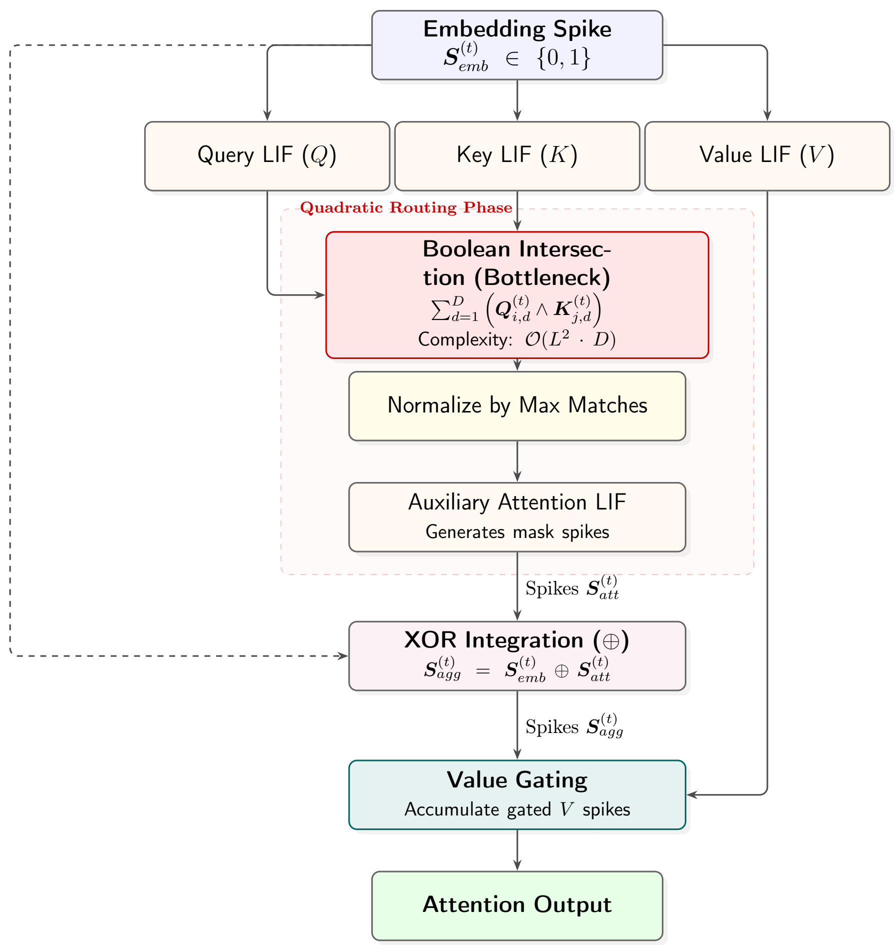

Figure 2: Detailed computational flow of the Spike-Overlap Attention mechanism at a single timestep.

temporal sequence T:

$$E = \sum_{t=1}^{T} \boldsymbol{V}_{pool}^{(t)} \tag{5}$$

Learning via BPTT and Add-Only Deltas: The true power of this architecture stems from its backpropagation mechanics. Because the Heaviside step function lacks a valid derivative, we employ a Rectangular Surrogate Mask to propagate errors, building upon the foundations of spatio-temporal backpropagation in SNNs [10]:

$$\sigma'\big(\boldsymbol{V}^{(t)}\big) = \begin{cases} 1 & \text{if } \left|\boldsymbol{V}^{(t)} - \theta\right| < w \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

where w represents the surrogate window width, dictating the range around the threshold θ where gradients are allowed to pass. During Backpropagation Through Time, the error $\delta^{(t)}$ at sequence step t is explicitly linked to future time steps via the heterogeneous decay $\boldsymbol{\beta}_{pool}$:

$$\tilde{\boldsymbol{\delta}}^{(t)} = \left(\boldsymbol{\delta}^{(t)}_{out} + \tilde{\boldsymbol{\delta}}^{(t+1)} \odot \boldsymbol{\beta}_{pool}\right) \odot \sigma'\big(\boldsymbol{V}^{(t)}_{pool}\big) \tag{7}$$

This explicit temporal unrolling allows the β parameter to act as a natural contextual router. The pooler inherently learns long-term syntactic dependencies simply by preserving or decaying gradients. Furthermore, during weight updating, the gradient accumulation $\Delta\boldsymbol{W}$ also utilizes an Add-Only Delta rule:

$$\Delta\boldsymbol{W}_{in,out} = \sum_{t=1}^{T} \tilde{\boldsymbol{\delta}}^{(t)}_{out} \cdot \mathbb{I}\left[S^{(t)}_{emb,in} = 1\right] \tag{8}$$

This greatly reduces dense matrix multiplications in the forward and BPTT routing phases, rendering the spatial $\mathcal{O}(L^2)$ matrix completely unnecessary. To perfectly preserve the sequence representation at initialization before learning temporal dynamics, the pooler is initialized as an Identity Matrix with biases disabled. We use a rectangular surrogate window width of $w = 1.0$.

2.4. Knowledge Distillation and Error Routing

To optimize these dynamics, we distill "dark knowledge" from a pre-trained dense Teacher, adapting established knowledge distillation techniques [11] to the discrete spiking domain. We utilize a Mean Squared Error (MSE) objective over the Pearson Correlation scores. The exact error $\mathcal{E}$ is given by:

$$\mathcal{E} = \mathrm{Sim}_{Teacher} - \sum_{d=1}^{D_{model}} \left(\tilde{\boldsymbol{E}}_{1,d} \cdot \tilde{\boldsymbol{E}}_{2,d}\right) \tag{9}$$

where $\tilde{\boldsymbol{E}}$ denotes the L2 unit-norm normalized, mean-centered pooled embeddings. The error gradient is derived symmetrically across the representation

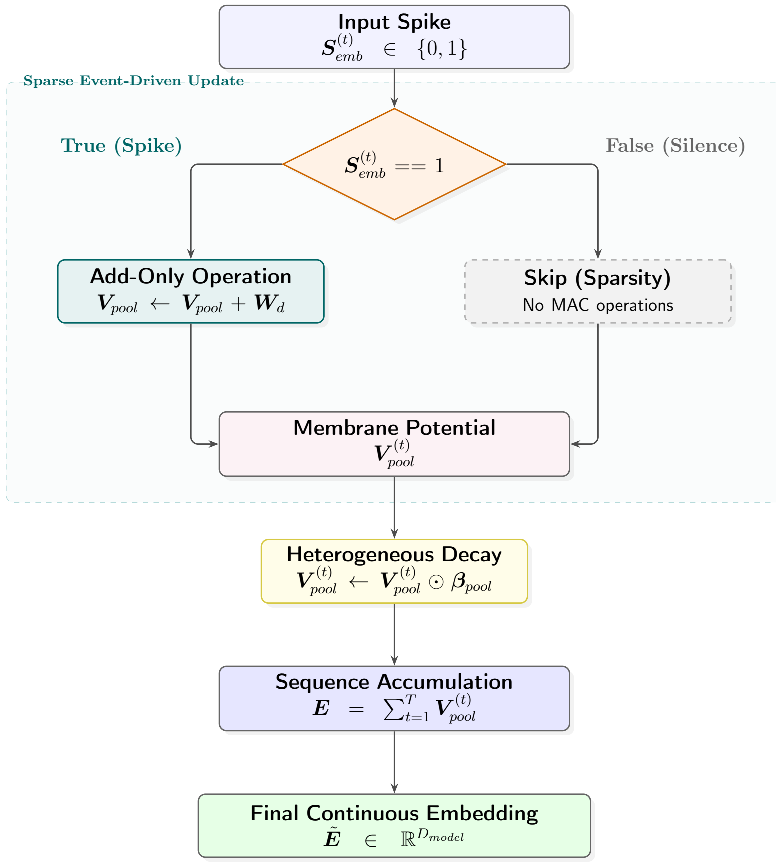

Figure 3: Internal computational flow of the purely recurrent Add-Only Pooler. Floating-point multiplications are completely eliminated.

space:

$$\frac{\partial \mathcal{L}}{\partial \tilde{\boldsymbol{E}}_{1,d}} = -\mathcal{E} \cdot \tilde{\boldsymbol{E}}_{2,d} \cdot m \tag{10}$$

$$\frac{\partial \mathcal{L}}{\partial \tilde{\boldsymbol{E}}_{2,d}} = -\mathcal{E} \cdot \tilde{\boldsymbol{E}}_{1,d} \cdot m \tag{11}$$

This gradient is then injected directly into the temporal unrolling sequence via BPTT. The contrastive margin $m = 0.5$ acts as a dampening factor that restricts explosive updates from sparse binary state flips. This end-to-end framework allows the discrete, event-driven parameter space to safely converge toward the continuous dense Teacher topology.

Table 1: Hardware Specifications for Training and Inference

Component	Specification
CPU Model	11th Gen Intel Core i5-1135G7
Base Clock Speed	2.40 GHz
System Memory (RAM)	16 GB
Operating System	Linux (x86_64)
Hardware Accelerator	None (Pure CPU Execution)

3. Experimental Evaluation

Hardware Specifications and Implementation: All models are implemented entirely in native Rust to guarantee deterministic execution and avoid the simulation overhead typical of Python-based frameworks like PyTorch. To highlight the extreme computational efficiency of the proposed SNN architecture, all training and inference evaluations were executed exclusively on a standard consumer-grade laptop without any GPU acceleration. The detailed hardware specifications are provided in Table 1.

Training Protocol: We trained over 10 epochs on a custom synthetic distillation dataset using a base learning rate of 2.0 with cosine annealing. For Knowledge Distillation, we construct this high-fidelity soft-label training dataset designed to map the continuous representation space of dense networks into the discrete binary domain. We extract over 250,000 synthetic sentence pairs from diverse textual corpora, maintaining strict separation to prevent data leakage with the constituent zero-shot evaluation benchmarks (STS-12 through STS-16, STS-B, and SICK-R). These pairs are passed through the pre-trained dense Transformer teacher model (`all-MiniLM-L6-v2`), which infers continuous dense vector embeddings for each sentence. We then calculate the Pearson Correlation between these dense vectors to generate absolute ground-truth correlation scores ranging from -1.0 to 1.0. The teacher model script and the dataset generation utilities are provided in the supplementary materials. Following training, the model is evaluated on the standard downstream tasks, where the aggregate ALL-STS score is calculated as a weighted average of Pearson correlations across these constituent benchmarks based on their respective sample sizes.

Table 2: Comprehensive Evaluation: Dimensionality Scaling and Architectural Ablation

Dataset / Metric	Recurrent ($D = 32$)	Recurrent ($D = 64$)	Recurrent ($D = 128$)	Recurrent ($D = 256$)	Attention ($D = 256$)
STS-12	0.3280	0.4166	0.4955	0.5343	0.5155
STS-13	0.3291	0.3106	0.4212	0.4472	0.4428
STS-14	0.3543	0.3676	0.4802	0.5083	0.4948
STS-15	0.3713	0.4188	0.5274	0.5228	0.5251
STS-16	0.3248	0.3145	0.3932	0.3648	0.4018
STS-B	0.6665	0.6656	0.7685	0.7651	0.7740
SICK-R	0.4477	0.4865	0.4838	0.4745	0.4819
ALL-STS (Pearson)	**0.6713**	**0.7092**	**0.8038**	**0.8031**	**0.8079**
Inference (ms/pair)	0.719	0.912	1.341	0.932	1.946
Train Time (seconds)	681	1,078	2,025	1,165	4,817
Avg Spikes/Sentence	840	1,668	3,358	2,157	6,737
Est. SNN SOPs (10^3)	17.3	79.5	339.2	1104.4	1407.4

Table 3: Dimensionality Saturation on Augmented Multilingual Data. It is important to note that the validation datasets evaluated here (Eval-Multi and All-STS-Teacher) reflect the augmented multilingual training distribution, which is distinct from the standard zero-shot STS benchmarks presented in Table 1.

Metric	Recurrent ($D = 256$)	Recurrent ($D = 512$)
Eval-Multi (Pearson)	**0.7297**	0.7233
All-STS-Teacher (Pearson)	**0.7478**	0.7469
Inference Latency (ms/pair)	**1.15**	1.91
Training Time (seconds)	**2,275**	4,319

3.1. Phase 1: Dimensionality Scaling and the Teacher Bottleneck

Using the Distillation strategy with the Recurrent Pooler, we scaled the embedding dimension D_{model} to analyze representation capacity. The SNN scales impeccably up to 256 dimensions, shattering the 0.80 Pearson correlation barrier, which is a massive improvement over prior Spike Coincidence Attention baselines [8] ($r = 0.758$).

However, we empirically observed a saturation point when scaling further to $D_{model} = 512$. As detailed in Table 3, in an expanded multilingual and synthetic typo-robustness evaluation, doubling the parameters from 256 to 512 resulted in a slight degradation of Pearson correlation (from 0.7297 to 0.7233 on Eval-Multi) while simultaneously nearly doubling the inference latency. We attribute this performance plateau to two primary factors: (1) **Teacher Bottleneck**: The dense teacher model projects continuous semantics into a $D = 384$ space. Distilling this compressed knowledge into a larger $D = 512$ discrete student architecture creates an inherent information bottleneck where the student possesses excess capacity without novel semantic signals to capture. (2) **Algorithmic Augmentation**: The synthetic negative sampling and typo perturbations utilized during training possess simple algorithmic properties. The $D = 256$ recurrent pooler fully encapsulates these patterns, leaving the $D = 512$ model highly susceptible to embedding space fragmentation and overfitting on hard negatives. Consequently, $D_{model} = 256$ serves as the optimal computational "sweet spot" for this specific SNN topology.

3.2. Phase 2: Ablation of Spike-Driven Attention

Fixing the dimension to $D_{model} = 256$, we performed an architectural ablation against the Spike-Overlap Attention and the minimalist Recurrent Pooler.

Discussion on Efficiency: The empirical data suggests that Spike-Driven Self-Attention provides marginal benefit relative to its computational cost for this specific sentence embedding context. While Spike-Overlap Attention yields a negligible raw accuracy bump ($\Delta = +0.0048$), it introduces a severe computational bottleneck. Incorporating the explicit $L \times L$ attention matrix (with $\sim 196K$ parameters compared to $\sim 65K$ for the Recurrent Pooler) effectively quadruples the total training time ($1,633$ seconds vs $4,817$ seconds) and doubles the execution latency (1.082 ms vs 1.946 ms) per sentence pair.

The Recurrent Pooler achieves computational superiority by bypassing spatial routing entirely. The BPTT gradients of the Recurrent Pooler's temporal context ($\beta \boldsymbol{V}_{pool}^{(t-1)}$) prove highly expressive, highlighting recurrent SNNs as a highly competitive and efficient topology for sequences under constrained budgets.

3.3. Phase 3: Generalization Across Semantic Benchmarks

To ensure the observed efficiency does not compromise zero-shot generalization, we evaluated the Recurrent Pooler against Spike-Overlap Attention across multiple standard benchmarks.

The Recurrent Pooler achieves highly competitive performance across all subsets, occasionally surpassing the Spike-Overlap Attention (e.g., STS-12, STS-13, STS-14). While attention occasionally improves performance on specific benchmarks (such as STS-16), its overall gains remain modest relative to the computational overhead incurred.

4. Limitations

While our attention-free recurrent SNN achieves remarkable sparsity, the peak correlation (0.8031) remains marginally below the theoretical limit of the full-precision Teacher model (> 0.85), indicating an inherent compression bottleneck in discrete binary signaling. Furthermore, our empirical sequence length is capped at $L = 128$; verifying temporal decay limits over much longer contexts (e.g., document-level embeddings) requires future investigation.

5. Conclusion

Through detailed mathematical formulation and rigorous low-level structural ablation, we demonstrate that forcing Spatial Attention onto Spiking Neural Networks produces an unjustifiable computational bottleneck. The biological temporal decay properties inherent to LIF neurons, when trained via Surrogate Gradients and BPTT, naturally encapsulate complex sentence semantics without requiring quadratic sequence routing. Our findings suggest that the benefits of spike-driven self-attention may not justify its computational cost in sentence embedding scenarios, paving the way for ultra-efficient, purely recurrent SNN embedders.

6. Code and Model Availability

To facilitate reproducibility and open science, the fully trained versions of our Attention-Free Spiking Sentence Embedder ($d_{model} = 256$) have been publicly released. The model weights, PyTorch-compatible SNN architecture scripts, and configuration files are available online on Hugging Face.

- **Base Model**: Achieves the reported $r = 0.8030$ Pearson correlation on the STS benchmarks. (*URL omitted for double-blind review. Code and weights will be released upon acceptance.*)
- **Multilingual Model**: Extends the vocabulary to $64,000$ to support cross-lingual semantic search across 20+ languages. (*URL omitted for double-blind review.*)

References

[1] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, Bert: Pre-training of deep bidirectional transformers for language understanding, arXiv preprint arXiv:1810.04805 (2018).

[2] N. Reimers, I. Gurevych, Sentence-bert: Sentence embeddings using siamese bert-networks, in: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), 2019, pp. 3982–3992.

[3] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need, Advances in neural information processing systems 30 (2017).

[4] Y. Tay, M. Dehghani, D. Bahri, D. Metzler, Efficient transformers: A survey, ACM Computing Surveys (CSUR) 55 (6) (2022) 1–28.

[5] K. Roy, A. Jaiswal, P. Panda, Towards spike-based machine intelligence with neuromorphic computing, Nature 575 (7784) (2019) 607–617.

[6] Y. Li, Y. Lei, X. Yang, Spikeformer: A novel architecture for training high-performance low-latency spiking neural network, arXiv preprint arXiv:2211.10686 (2022).

[7] R.-J. Zhu, Q. Zhao, G. Zhang, P. Deng, H. Li, Y. Duan, X. Cheng, et al., Spikegpt: Generative pre-trained language model with spiking neural networks, arXiv preprint arXiv:2302.13939 (2023).

[8] Anonymous, Details omitted for double-blind reviewing, uRL omitted for double-blind review (2026).

[9] W. Fang, Y. Chen, J. Ding, D. Chen, Z. Yu, H. Zhou, M. Timothée, Y. Tian, et al., Spikingjelly: An open-source machine learning infrastructure platform for spike-based intelligence, Science Advances 9 (40) (2023) eadi1480.

[10] E. O. Neftci, H. Mostafa, F. Zenke, Surrogate gradient learning in spiking neural networks: Bringing the power of gradient-based optimization to spiking neural networks, IEEE Signal Processing Magazine 36 (6) (2019) 51–63.

[11] G. Hinton, O. Vinyals, J. Dean, Distilling the knowledge in a neural network, arXiv preprint arXiv:1503.02531 (2015).